

Setting Standards for Stablecoin Reserves

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Abstract

US regulators have suggested that only FDIC-insured financial institutions be allowed to issue stablecoins. We propose an alternative path, which provides similar guarantees in terms of financial resilience, but without the same narrow restriction on issuers and negative consequences on innovation and competition. Stablecoin issuers should comply with the capital and liquidity standards encoded in the Basel accords, and put aside appropriate capital buffers to mitigate credit risk, market risk, and operational risk. Furthermore, stablecoin issuers should hold appropriate liquidity to mitigate sudden redemptions and outflows. While many existing stablecoin issuers would struggle to comply with such standards under their current reserve design, they could successfully comply under a simpler balance sheet centered around short-maturity, high-quality, and liquid assets such as 3 months or less U.S. Treasuries.

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1 Introduction

Worth under \$10 million just five years ago, stablecoins are suddenly a mainstream asset class, with a market capitalization well over \$100 billion today.¹ But this new prominence brings increased scrutiny from policymakers – most recently from the President’s Working Group (2021). In addition to well known illicit finance concerns, the President’s Working Group identified three key dimensions of risk for stablecoins: loss in value of the coins, failures in the payments system, and amplified risks for large-scale systems. Focusing on the first two risks, they recommend that only banks issue stablecoins:

*Legislation should limit stablecoin issuance, and related activities of redemption and maintenance of reserve assets, to entities that are insured depository institutions. The legislation would prohibit other entities from issuing payment stablecoins.*²

At face value, this recommendation makes sense. While stablecoin issuers point to price histories or high-level audits to build confidence, this is unlikely to be enough as coins move beyond crypto use cases and are used for mainstream payments. As the history of financial institutions painfully reminds us, balance sheets need to be able to withstand financial distress, operational failures, and more to foster trust.

At the same time, limiting the issuance of stablecoins to banks also has significant side-effects. It limits competition and innovation to incumbents, and decreases the likelihood that consumers and businesses will benefit from lower costs and new types of products and services.

¹Values are taken from “Total Stablecoin Supply,” published by The Block; and accessed on November 11, 2021.

²The President’s Working Group further clarifies: “Insured depository institutions include both state and federally chartered banks and savings associations, the deposits of which are covered, subject to legal limits, by deposit insurance, and which have access to emergency liquidity and Federal Reserve services.”

But there is another path that can both minimize risk and unlock innovation: stablecoin issuers should follow established bank standards for their reserves. More specifically, regulators could build on the guidelines developed by the Basel Committee on Banking Supervision (“Basel accords”) to harmonize reserve designs. These standards have been developed and improved over the last fifty years to keep financial institutions solvent and liquid through a variety of distressed scenarios. With some adjustments and extensions, they therefore provide a rigorous and effective approach to regulating stablecoins.

2 A Short Primer on Capital and Liquidity Requirements

The Basel accords focus on **capital** and **liquidity requirements**, which are essential for banks to remain solvent and liquid through distress.³ Consider first capital requirements, which focus on defining an appropriate **capital buffer** for a bank’s balance sheet. Conceptually, a capital buffer refers to extra assets on the balance sheet that are not associated with liabilities. Such a buffer is critical to absorb unexpected losses.⁴

Take a simple example: a bank that collects \$100 in cash from the public, in exchange for \$100 in deposits. This bank has no excess assets on its balance sheet, and

³As some history, the committee that drafts the accords, the Basel Committee on Banking Supervision, was first established in 1974 following the failure of Bankhaus Herstatt in West Germany. Over the subsequent fifty years, the committee evolved, growing from a coalition of ten to twenty-eight nations, while continuously publishing enhancements to supervision. On a few occasions, the committee synthesized and codified these into a comprehensive set of standards. Following the Latin American debt crisis in the 1980s, the committee released the first full set of banking standards known as Basel I in 1988; and this was followed by Basel II in 2004. Following the Financial Crisis of 2007 - 2009, the committee released the current version of standards, known as Basel III in 2010; and a fourth refinement (known as Basel IV) is slated to be released in 2023.

⁴Note that, per this definition, capital buffers exist in every business in the world mechanically. A business that has fewer assets than liabilities is bankrupt; and so all non-bankrupt businesses have some implicit capital buffer.

thus no capital buffer. Should the bank lose \$1 of assets (due to, say, theft), the bank would hold fewer assets than liabilities, become insolvent and immediately be vulnerable to a run.

Assets	Liabilities		Assets	Liabilities
\$100 Cash	\$100 Deposits	→	\$99 Cash	\$100 Deposits

Now, suppose the same bank had raised \$2 from its shareholders without issuing them deposit liabilities. In this scenario, the bank has \$2 in excess, or \$2 capital buffer.⁵ Should the bank lose \$1 in assets, it would still hold more assets than liabilities and remain solvent. This illustrates the critical role of a capital buffer.

Assets	Liabilities		Assets	Liabilities
\$102 Cash	\$100 Deposits	→	\$101 Cash	\$100 Deposits
	\$2 Capital			\$1 Capital

Next, consider **liquidity requirements**, which focus on ensuring an institution can meet unexpected redemptions of its deposits. A bank that holds a large share of illiquid assets would be unable to redeem a sizable share of its deposits. To illustrate, consider a bank with no capital buffer that takes \$100 in cash in exchange for \$100 in deposit slips, and then invests the \$100 in residential mortgages. Should depositors redeem unexpectedly, the bank will be forced to sell some of its mortgages quickly, but since mortgages are hard-to-value assets, it will likely not get fair market price and

⁵The capital buffer does not refer to some segmented portion of the balance sheet, nor do the assets need to be held in some special form – it is just an accounting concept to describe the overcollateralization of a balance sheet. In this scenario, the \$2 need not be held in a separate vault, nor do they necessarily need to be held as cash.

have to sell the mortgages at a discount. This would make the bank insolvent and immediately vulnerable to a run.

Assets	Liabilities		Assets	Liabilities
\$100 Mortgages	\$100 Deposits	→	\$1 Cash \$99 Mortgages	\$100 Deposits

Now, suppose the bank purchased mortgages, but also held an appropriate part of its balance sheet in cash. Cash offers easy access to liquidity to meet redemptions, and helps the bank meet its obligations without triggering insolvency. This illustrates the importance of liquidity standards.

Assets	Liabilities		Assets	Liabilities
\$10 Cash \$90 Mortgages	\$100 Deposits	→	\$9 Cash \$90 Mortgages	\$99 Deposits

3 Applying the Basel Framework to Stablecoins

The Basel accords are primarily focused on ensuring solvency and that financial institutions have enough capital to hedge risks. These risks are the same stablecoin issuers face: **credit risk**, **market risk**, and **operational risk**. The guidance is risk-sensitive, meaning that larger risks need to be offset by larger capital requirements. Riskier assets on a balance sheet or, similarly, an increase in the riskiness of the existing assets need to be met with a larger capital buffer.

The first category of risk is **credit risk**, or the risk that the backing assets will partially or fully default. For each dollar of assets on a balance sheet, the accords prescribe some amount of capital to be held against the risk of default.

The following table illustrates requirements for a range of assets (normalized as required capital per \$100 of underlying assets).⁶ Many of these assets are commonly held by existing stablecoin issuers. As evident from the table, riskier assets such as commercial paper require substantially more capital than creditworthy assets such as Treasury bonds.

Underlying Asset	Credit Risk-Based Capital Per \$100
Treasury bonds	\$0.00
Deposits with the Federal Reserve	\$0.00
Deposits with highly-rated bank ⁷	\$1.60
Commercial paper rated A-1/P-1 ⁸	\$1.60
Commercial paper rated A-2/P-2	\$4.00
Commercial paper rated A-3/P-3	\$8.00
Bitcoin and other cryptocurrencies ⁹	\$100.00

⁶This table assumes an 8% minimum capital ratio, in converting risk-weighted assets into capital requirements. Note that this is only one approach to valuing credit risk: the Basel accords offer several approaches, and national regulators further have substantial discretion in modifying these approaches.

⁷“Highly-rated” refers to banks that have earned general credit ratings of AA- (SP) or Aa3 (Moody’s) or higher. These are further documented in the Basel Framework’s CRE20.

⁸These refer to short-term commercial paper issues that are directly rated, per the notation used by SP and Moody’s, as documented in the Basel Framework’s CRE21. If the issues are neither short-term nor directly rated, then the rules from CRE20 regarding exposures by banks and corporates would apply.

⁹The Basel Committee on Banking Supervision has not finalized a framework for cryptocurrencies, but this comes from their consultative document “Prudential treatment of cryptoasset exposures” (dated September 10, 2021). This document distinguishes between three classes of assets: tokenized traditional assets, cryptoassets with stabilization mechanisms (e.g. stablecoins), and all other cryptoassets.

Stablecoin issuers that hold riskier assets without a corresponding increase in capital buffer are at risk of becoming insolvent and exposed to a run.

The second category of risk is **market risk**, or the risk that the market price of the backing assets will fall. For each dollar of assets, Basel prescribes some amount of capital to hold against market shocks, with larger requirements associated with more volatile assets.

Stablecoin issuers often hold fixed-rate assets such as commercial paper or Treasury bonds. As a result, their core market risk is interest rate risk, or the risk that interest rates will rise and the assets will fall in price. The following table illustrates requirements for different maturities of fixed-rate assets (normalized as required capital per \$100 of the underlying assets).¹⁰

Underlying Maturity	Interest Risk-Based Capital Per \$100
0-1 Months	\$0.12
1-3 Months	\$0.48
3-6 Months	\$1.03
6-12 Months	\$1.84
12-36 Months	\$3.90

Stablecoin issuers that hold longer maturity assets without increasing their capital buffer, are at greater risk of insolvency and a run.

¹⁰Note that this is only one approach to valuing market risk, known as the Interest Rate in the Banking Book approach (documented in the Basel Framework's SRP31). There are alternate approaches, including the Standardized Approach (documented in the Basel Framework's MAR10 and MAR20) and the Internal Models Approach (documented in the Basel Framework's MAR10 and MAR30). In practice, national regulators will work with financial institutions to identify the appropriate approach.

The third category of risk is **operational risk**, or the risk that internal processes, people, or systems will fail. This serves as a catch-all bucket for operational failures, such as wrongful termination lawsuits, theft, government fines, and much more. While the Basel accords include risks typically encountered by banks, these risks do not always translate to stablecoins.¹¹ Stablecoins issuers may need more bespoke approaches that cover novel categories of risk such as bugs in smart contracts, problems with network validators, unauthorized or impeded minting and burning operations, vulnerabilities in cryptographic primitives, financial crime risks and more.

The Basel framework also offers guidance on aggregating categories of risk, and introduces the concept of a “leverage ratio”, i.e. a minimum capital requirement that is completely risk-insensitive and directly grows with the size of a balance sheet. This type of ratio was designed for institutions that have complex balance sheets, and is unsuited for stablecoin issuers that do not engage in extensive maturity transformation and only invest in short term U.S. Treasuries.¹² If applied to stablecoins, it could make them commercially non-viable.¹³

To identify unexpected weaknesses in a balance sheet, the Basel framework relies on stress tests which typically include some combination of credit, market, and operational shocks, as well other shocks to the underlying business or economy. In the case of stablecoins, for example, stress tests reveal that sudden expansions in supply can be

¹¹To compute operational risk capital, the Basel framework captures three high-level statistics regarding a bank’s balance sheet: its interest and dividend flows, its operating and fee income and expenses, and its balance sheet’s profit and loss. It then maps these exposures into operational risk-based capital based on the bank’s aggregate exposures and its historical success in mitigating operational risk events. Further details are available in the Basel Framework’s OPE25, scheduled to come into effect in January 2022.

¹²There are further forms of aggregation, but they matter less for stablecoin issuers, given the different role they play in the economy as compared to banks. For instance, these requirements will be layered between a base Pillar 1 requirement and an additional Pillar 2 requirement (where banks that breach the Pillar 2 requirement face restrictions on dividends and bonuses, but banks that also breach the Pillar 1 requirement can be wound down). The Basel framework further offers recommendations regarding countercyclical capital buffers, additional buffers for globally systemic financial institutions, and more.

¹³See Catalini and Massari (2021).

as problematic as contractions, as they require issuers to raise capital quickly to grow their capital buffer. While stopping minting may seem an appealing way to slow down growth, it would be a disastrous one as it would lead to the stablecoin breaking its peg upwards.¹⁴

The Basel accords also focus on ensuring that financial institutions have enough liquidity to meet unexpected outflows. They rely on two key metrics to measure liquidity, one of which is highly relevant to stablecoin issuers. This metric is the liquidity coverage ratio, which ensures that banks have enough “high-quality, liquid assets” (HQLAs), i.e. assets that can be easily sold for cash to meet prolonged and extensive redemptions. This prevents a bank from having to sell less liquid assets at a discount, and potentially being unable to meet redemptions altogether. Formally, the ratio is computed as the stock of HQLAs divided by the net cash outflows over the next thirty days, and it should always exceed 100%.¹⁵ The framework also computes a worst-case scenario for net cash outflows, where a bank’s creditors who are considered flighty (e.g. institutional investors) raise the net cash outflow totals more than creditors who are considered sticky (e.g. retail depositors). Finally, cash that a bank holds in other banks’ accounts qualifies as a special category: rather than adding to the stock of HQLAs, it can lower the net cash outflows – but only up to a point.¹⁶

¹⁴In fact, a sudden contraction in the balance sheet can often be easier for stablecoin issuers to handle, particularly if their balance sheets are highly liquid. This is because assets can be liquidated very easily and with very small losses, and outflows thus free up capital that can be used to stabilize the remainder of the issuer’s balance sheet.

¹⁵There are several tiers of HQLAs, where higher ranked “Tier 1” HQLAs can be used without restriction, while lower ranked “Tier 2” HQLAs face restrictions. Most government securities, including Treasury bonds, qualify as Tier 1 assets. Highly-rated commercial paper qualifies as a Tier 2A asset, and less highly-rated non-financial assets qualify as Tier 2B assets.

¹⁶In assessing the stock of HQLAs, Level 1 assets – which include central bank reserves, sovereign bonds, and physical cash – can be counted without restriction. However, Level 2 assets both take a haircut in value and in total can only count for 40% of the stock of HQLAs. In assessing net cash outflows, all liabilities on a bank’s balance sheet are multiplied by a factor known as a run rate and summed, and cash positions on the balance sheet can decrease this by up to 75% of its original value. Stablecoin deposits are not explicitly covered by the Basel framework, but there are several similar options. In particular, demand deposits to individuals that do not benefit from FDIC insurance have run rates of 10%, while unsecured wholesale financing provided to miscellaneous institutions have

4 Towards an Ideal Reserve Design

The framework outlined so far can not only be applied to existing stablecoins, but also provides guidance on what an ideal reserve balance sheet should look like.

For instance, consider Tether, who released a breakdown of its reserves as of June 30, 2021.¹⁷ Tether had approximately \$63 billion in assets, of which they provided sufficient detail on some \$53.4 billion to estimate capital and liquidity requirements. This portion of Tether's balance sheet consisted of cash and bank deposits, commercial paper, short-maturity Treasury bonds, and reverse repos.

In total, these generate estimated capital requirements amounting to 2.9% of Tether's balance sheet (excluding additional capital needed to cover operational risk), and they generate an estimated liquidity coverage ratio between 58% and over 100%, depending on exactly how flighty Tether's liabilities actually are.¹⁸ It is likely that the estimated

run rates of 100%. Further details can be found in the Basel Framework's document "Basel III: the Liquidity Coverage Ratio and liquidity risk monitoring tools" (dated January 2013). The Basel accords also compute a second metric, known as the net stable funding ratio, to measure liquidity. This ensures that banks have a sustainable funding profile for their assets, e.g. a bank isn't forced to find new depositors to substitute for old depositors when holding long-term and encumbered assets. As most stablecoin issuers hold marketable and unencumbered securities, this metric is not particularly relevant. The net stable funding ratio computes the ratio of available amount of stable funding to the required amount of stable funding, and mandates the ratio should exceed 100%. To compute available stable funding, each liability on the balance sheet is multiplied by an ASF factor and summed; and to compute required stable funding, each asset on the balance sheet is multiplied by a RSF factor and summed. Demand deposits, which map most closely to stablecoins, qualify for ASF factors between 50 - 95%, while Treasuries, commercial paper, and other marketable securities qualify for RSF factors between 0 - 50%. Thus, any solvent stablecoin issuer will satisfy this ratio – although issuers that venture into more exotic assets, such as residential mortgages, may want to watch this more closely. Further details can be found in the Basel Framework's document "Basel III: the net stable funding ratio" (dated October 2014).

¹⁷https://tether.to/wp-content/uploads/2021/08/tether_assuranceconsolidated_reserves_report_2021-06-30.pdf

¹⁸This calculation makes several assumptions, as Tether does not provide sufficient granularity to compute these exactly, and regulators have not issued guidance on the appropriate regulatory treatment for stablecoins. Tether provides a compositional breakdown of its assets: cash and bank deposits (by assumption, held at highly-rated banks) worth \$6.3 billion, A-1 commercial paper and CDs worth \$14.7 billion, A-2 paper and CDs (including reverse repos) worth \$15.0 billion, A-3 paper and CDs (including, by assumption, assets marked as "other") worth \$2.2 billion, and Treasuries worth \$15.3 billion. This generates a credit risk-based capital requirement of \$1.1 billion. Tether also provides a maturity breakdown of its assets: \$26.9 billion between 0 - 90 days, inclusive of its Treasury and (by assumption) reverse repo positions, \$6.5 billion between 90 - 180 days, and \$13.7

capital requirements and liquidity coverage ratio respectively rise and fall when expanded to Tether's entire balance sheet, as the remaining portion of the balance sheet includes secured loans, precious metals, cryptocurrencies, and other assets that are substantially riskier and less liquid.

Alternatively, consider USDC, who released a breakdown of its reserves as of September 30, 2021.¹⁹ USDC had \$31.7 billion in assets, all held as cash and cash equivalents. In total, these generate estimated capital requirements under the Basel framework amounting to 1.6% of Circle's balance sheet (again excluding additional capital needed to cover potential operational risk losses), and generate an estimated liquidity coverage ratio of 0%.²⁰

A simpler solution, originally proposed by Diem,²¹ is to hold a balance sheet that is almost entirely short-term Treasury bills, with a small allocation to cash. For instance, suppose a balance sheet that is held 90% in Treasury bills with a residual maturity of under three months, and 10% in cash at a well-rated financial institution. This balance sheet would have a capital requirement that is under 0.5% of its balance sheet (holding aside operational risk-based capital) and a liquidity coverage ratio that is always over 100%, regardless of the treatment that regulators apply to stablecoin liabilities.

billion between 180 - 365 days. This generates a market risk-based capital requirement of \$0.4 billion. Finally, in computing the liquidity coverage ratio, Tether can utilize its stock of Treasuries and reverse repos as Tier 1 assets, and (by assumption) can use its stock of commercial paper as Tier 2 assets, which are capped at 40% of the total stock of HQLAs. Tether can further use its entire cash position to reduce its net cash outflows.

¹⁹<https://www.centre.io/hubfs/pdfs/attestation/2021%20Circle%20Examination%20Report%20September%202021%20Final.pdf>

²⁰This calculation makes several assumptions, as Circle does not provide sufficient granularity to compute these exactly. Circle notes that the entirety of its \$31.7 billion reserve position is held as cash and cash equivalents (for simplicity, assumed to be entirely cash held at highly-rated banks). This generates a credit risk-based capital requirement of \$0.5 billion; and there is no market risk-based capital requirement. Finally, in computing the liquidity coverage ratio, Circle can use its cash assets to offset its net cash outflows by 75% of the original value; but it has no HQLAs (e.g. Treasuries) for the numerator of the calculation.

²¹<https://www.diem.com/>

This makes for a safer reserve, reducing the capital and liquidity burdens on the issuer, while ensuring that the underlying coin can withstand severe distress.

While at face value, the simplest solution may seem for a stablecoin issuer to back a stablecoin with cash at one or more commercial banks, this introduces a different set of challenges. Cash at a bank has no interest rate risk, appears fully liquid, and benefits from protections by the Federal Deposit Insurance Corporation (FDIC). At the same time, there are three shortcomings to an all-cash balance sheet for a stablecoin.

First, while consumers are protected by FDIC, institutions are not. As such, if a bank goes bankrupt, a stablecoin's cash deposit may be lost.²² Second, banks do not want to hold a large amount of cash because it comes with large capital requirements, and in a low interest rate environment can be completely uneconomical.²³ Third, cash may not be as liquid as it appears. The liquidity standards in the Basel accords do not allow for unrestricted usage of cash to meet the liquidity coverage ratio, as stablecoin issuers run the risk that a bank may suspend access to its deposits – particularly if the bank is undergoing the same liquidity crunch that the stablecoin issuer is facing.

This calculus would rapidly change if stablecoin issuers could hold deposits directly with the central bank. Cash held with the central bank has no credit risk, and unlike private banks, the central banks typically do not limit the amount of cash that financial institutions hold with them, nor suspend access to those deposits. Indeed, such a reserve would be almost riskless, and stablecoin issuers would only have to hold a capital buffer to cover operational risks. However, while central banks are exploring

²²The FDIC provides \$250,000 in insurance to accounts, which is too low for institutional clients like stablecoin issuers. Note that the FDIC does have a provision called pass-through insurance, in which funds deposited at an FDIC-insured bank by a fiduciary on behalf of multiple owners will qualify for FDIC insurance as if the actual owners had made the deposits individually. However, the general applicability of pass-through insurance to stablecoins is unclear, and only a handful of stablecoins (e.g. Gemini USD) have secured pass-through insurance for a subset of their depositors.

²³The binding constraint tends to be the leverage ratio, in which banks must hold at least 3% of their assets as a capital buffer (and 5% for globally systemically important banks), even when the assets are low-risk and low-yielding, like cash.

ways to support central bank digital currencies and expand access to fintech players, stablecoin issuers do not have access to central banks today. As a result, the best design remains one that relies on high quality, liquid assets, such as short term U.S. Treasuries.

While the President's Working Group has correctly identified issues with the status quo, the cure should not restrict innovation to incumbents. The Basel accords offer a number of ideas on how to design safer stablecoins that can withstand stressed market conditions. Expanding the framework and using it for a new type of bank charter that takes a "same risks, same rules" approach but also does not block entry by new players may be a more promising avenue for supporting competition and innovation in payments and financial services.

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